

Hermes[**@T.I**]

// effective_system = agent._cached_system_prompt (+ ephemeral), sent as `api\_messages = [{"role": "system", ...}] + api\_messages`. Everything below is concatenated into that single system message.

ROLE: SYSTEM

Frag **SystemPrompt**[**@T**]

// — PREFILL TURNS —

// Config-supplied conversation turns (prefill_messages_file), inserted directly AFTER the system message and BEFORE any history. Each entry carries its own role, so the rendered roles are data-dependent.

Frag **PrefillMessages**[**@T**]

// — COMPACTION HANDOFF —

// Compaction replaces the middle of the transcript with one summary row. It fires at turn start, after an idle gap and after a tool round. NOTE: the cut is a MESSAGE-row boundary (protect_first_n head rows + a token-budgeted tail), not a turn boundary – @C below is the turn that boundary falls in, which is an approximation of a row-level cut.

// none when no compaction has yet occurred, or when a few compactions already occured

◇ If **sys.number_of_compactions**[**@T**] = 1:

Frag **ProtectedFirstMessages**[**@T**]

// The summary row's role is chosen so user/assistant alternation stays valid on both sides of it.

Frag **CompactionSummary**[**@T**]

// — REPLAYED HISTORY (completed turns) —

// A turn is one run_conversation() call: one inbound ask, a run of sub-steps, and a closing row that may be absent. User-role rows the loop appends mid-turn (continuation prompts, /steer corrections) belong to the same turn.

// The code builds the request from one flat message list; turns, sub-steps and their kinds are read off row patterns here for readability.

Name **C** := **sys.last_compaction_turn**[**@T**] *// 0 while no compaction has occurred*

// — remaining replayed turns —

↻ **ForEach** **t** : **@C + 1...@T**

ROLE: USER

◇ If **@t** = **@C + 1**:

Frag **FirstTurnAfterCompaction**[**@t**]

◇ Else:

Frag **InputMemoriesAndPlugins**[**@t**]

// one HistoryStep per model response that did not close the turn; only the permanent kinds (tool_round, continuation, user_correction) survive into a completed turn

↻ **ForEach** **i** : **@t.substeps**

Frag **HistoryStep**[**@t, i**]

// absent when the turn was stopped before any model reply; the next ask then follows directly and the two user rows are merged on the wire

◇ If **sys.turn_has_closing_reply**[**@t**]:

Frag **TurnClosingReply**[**@t**]

// — CURRENT TURN —

// the inbound ask, composed live; the composed bytes are stamped on the row and are what Mark 4 replays on later turns

ROLE: USER

Frag **InputMemoriesAndPlugins**[**@T**]

↻ **ForEach** **i** : **@T.substeps[: -1]**

Frag **HistoryStep**[**@T, i**]